EXAMPLES OF QUANTUM MULTIPLICATION

(a) The flag variety $Fl(1,2,3)$.

(b) The complete intersection of $\mathbb{P}^2 \times \mathbb{P}^2$ by $\mathcal{O}(1,1)$ should give the same result as $Fl(1,2,3)$.

(c) $\mathbb{P}^4$ with the line bundle $\mathcal{O}(4)$.

(d) The complete intersection of $\mathbb{P}^3 \times \mathbb{P}^3$ by the bundle $\mathcal{O}(1,1) \oplus \mathcal{O}(1,1)$. This should be the same as the complete intersection of $Fl(1,3,4)$ by the line bundle $\mathcal{O}(1,1)$.

(e) Consider the flag variety $SO(5)/B$. This is the root system of type B_2 with roots $\{1,2\}$, and the complete intersection is defined by the line bundle $\mathcal{O}(1,1)$.

(f) The complete intersection of $\mathbb{P}^3 \times \mathbb{P}^3$ by the bundle $\mathcal{O}(1,1) \oplus \mathcal{O}(1,1) \oplus \mathcal{O}(1,1)$.

(g) The complete intersection of $Gr(2,5)$ by the bundle $\mathcal{O}(1) \oplus \mathcal{O}(1) \oplus \mathcal{O}(2)$.

(h) The complete intersection of $\mathbb{P}^3 \times \mathbb{P}^3$ by the bundle $\mathcal{O}(1,1) \oplus \mathcal{O}(2,2)$.

(i) The complete intersection of $Fl(1,2,5)$ by the bundle $\mathcal{O}(0,1) \oplus \mathcal{O}(0,2) \oplus \mathcal{O}(1,0)$.

(j) Complete intersection of $\mathbb{P}^1 \times \mathbb{P}^5$ by the bundle $\mathcal{O}(1,1) \oplus \mathcal{O}(0,3)$

(k) Complete intersection of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^4$ by the bundle $\mathcal{O}(1,1,1) \oplus \mathcal{O}(0,0,3$